

Optical pulling forces for 3-dimensional motion control in plasmonic nanomotors

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Abstract: We present a novel nanomotor design that enables 3-dimensional control of motion. A moderate cone angle Bessel beam with azimuthal polarization, acting on a glass chassis, allows for stable optical pulling; while asymmetric plasmonic dimers embedded within this chassis provide lateral movements upon plane wave illumination.

Momentum transfer in light-matter interactions generates optical forces, which play a crucial role in advancing micro- and nanotechnology, particularly in lab-on-a-chip devices [1]. Traditionally, gradient optical forces from optical tweezers have been employed to drive components such as pumps and valves, but these forces are limited by the size, shape, and material properties of the actuated objects. To overcome these limitations, recent research has proposed nanomotors utilizing scattering forces from asymmetric plasmonic nanoantennas or dielectric metasurfaces [2,3]. While showing successful results in enabling movement within two-dimensional planes, controlling longitudinal motion has remained a persistent challenge [4]. In contrast to the relatively trivial and well-known pushing effect obtained from scattering, optical pulling forces, first recognized in the last decade, require precise illuminating and scattering conditions [5].

This optical pulling effect has been theoretically achieved by the exploitation of several schemes, such as large cone angle Bessel beams, plane wave pairs, hyperbolic materials or chiral light-matter interactions [6]. However, most of these strategies propose unrealistic experimental conditions, making the practical realization of the effect challenging. A promising solution to this problem was proposed by Li et al. in 2019, using a dielectric cylinder that collimates Bessel beams by acting as a waveguide, allowing optical pulling effects across long distances with low cone angle Bessel beams [7].

In this work, we present a novel nanomotor design that enables control over both transversal and longitudinal motion. The system utilizes a coupling mechanism between an azimuthally polarized Bessel beam and a dielectric glass cylinder, which serves also as a chassis for the nanomotor (Figure 1a). Asymmetric plasmonic dimers are embedded within the cylinder in a way that minimizes their interaction with the pulling beam. This scheme allows for lateral motion, driven by the scattering forces from the plasmonic dimers under plane wave illumination (Figure 1b,d), while Bessel beam incidence enables longitudinal motion through optical pulling forces on the cylinder. Our design unlocks a new degree of freedom in motion control, allowing for pulling, pushing, and lateral movement by adjusting the polarization or switching between plane waves and Bessel beams. Furthermore, the optical pulling force in our system is robust against rotational and translational disturbances (Figure 1c), including Brownian motion, enhancing its practical applicability for advanced optical manipulation applications.

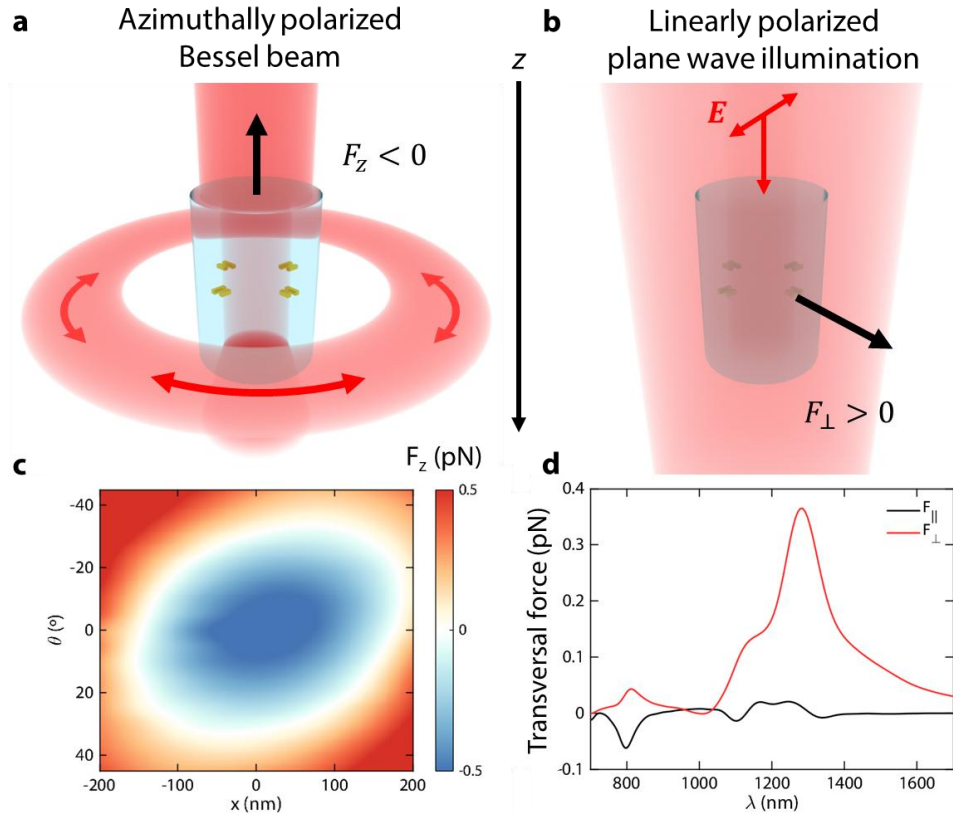


Figure 1. Depiction of the nanomotor design, acting in its optical pulling (a) and lateral movement (b) configurations. c) Longitudinal optical force with respect to angle and position. d) Lateral force components with respect to incident wavelength.

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